25900012

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Multi-Disciplinary Courses (MDC) - MG2MDCBCH100 - BIOCHEMISTRY IN ENTREPRENEURSHIP (2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember (K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall
outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries 1 marks

1. Long-termunhealthylifestylechoicesinanindividualleadsto [K]/[CO1] **ANS d).Lifestyledisorders**

2. How can a student interested in biochemical entrepreneurship enhance their skills and knowledge? [U] [CO5]
ANS a). Participating in internships, competitions, and research related to biochemistry and entrepreneurship

3. Select which regulatory body oversees dietary supplement in the India?[E] / [CO2]
ANS a)FSSAI

4. Examine which marketing activity for earned media coverage instead of direct advertising [K] / [CO4]
ANS: a) Public relations

5. Omega-3 fatty acids are commonly found in [K] / [CO2]
ANS a). Fish oil supplement

Part B
Fill In The Blank Questions
Answer all questions
Each question carries 2 marks

6. _____ is the main function of carbohydrates in the diet. [U] / [CO1]
ANS: Providing energy

7. The major classifications of traditional nutraceuticals include dietary supplements, _____, and probiotics. [K] / [CO2]
ANS: Herbal Products
8. Marketers design strategies to communicate product benefits effectively to _____ [A] / [CO5]

ANS : target consumers

9. Name two imaging techniques utilize biochemical principles. [[K] / [CO5]
   ANS: computed tomography (CT), positron emission tomography (PET), and magnetic resonance imaging(MRI) (Any two)

10. Cleanroom environments are used to minimize the risk of _____ [U] / [CO3]
    ANS: contamination.

Part C
Short Answer Questions
Answer any 5 questions
Each question carries 3 marks

11. What is the role of Omega-3 fatty acids in heart health? [K] / [CO1]
    ANS: Any three points (one mark each)

12. How do nutraceuticals differ from regular food products? [U] / [CO2]
    ANS: Any three differences (One mark each)

13. . Name two antioxidant-rich nutraceutical products and explain their benefits. [U] / [CO5]
    ANS : Name any two antioxidants and their one/two benefits (1.5 mark each)

14. Why might two nutraceutical products with similar ingredients perform differently in the market? [An] / [CO3]
    ANS: Explain any three factors (one mark each)

15. What role do probiotics and prebiotics play in the nutraceutical industry [A] / [CO5]
    ANS: Role of prebiotics (1.5 Marks)
    Role of Probiotics (1.5 Marks)
16. What is the projected market size of the global nutraceutical industry by 2030? [U] / [CO5]
    ANS: Explain the market size of the global nutraceutical industry (3marks)

17. Write a note on phytonutrients. [U] / [CO2]
    ANS: Explain any three phytochemicals (one mark each)

Part D
Short Essay Questions
Answer any 1 questions
Each question carries 5 marks

18. Write an essay on herbal supplements [U] / [CO1]
    ANS: Name and Explain any 5 herbal supplements (0ne mark each)
19. Describe FDA approval process in detail. Explain labelling provisions for nutraceutical under FSSAI. [U] / [CO4]
    ANS: Explain FDA approval process (2.5 Marks)
    Explain labelling provisions for nutraceutical underFSSAI. **(2.5 Marks)**